AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2024

Business Statistics Paper II

Time: 1 hour 5 minutes Marks: 20

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are SIX questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.

6. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols have their usual meanings.

Measures of Central Tendency and Quartiles

$$\overline{X} = \frac{\sum wx}{\sum w} \qquad \overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{\sum f} \qquad \mathrm{Q}_i = l + \frac{h}{f}\left(\frac{n \times i}{4} - c\right)$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Measures of Dispersion

$$\text{variance} = \frac{\sum\left(x - \overline{X}\right)^2}{n} \qquad \text{SD} = \sqrt{\frac{\sum\left(x - \overline{X}\right)^2}{n}}$$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{SD} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{variance} = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2 \qquad SD = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

$$\text{variance} = \frac{\sum f\left(x - \bar{x}\right)^2}{\sum f} \qquad SD = \sqrt{\frac{\sum f\left(x - \bar{x}\right)^2}{\sum f}}$$

$$\mathrm{R} = x_{\max} - x_{\min} \qquad \text{Coefficient of Variance} = \frac{SD}{\overline{X}} \times 100$$

Index Number

$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100 \qquad P_{0n} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times 100 \qquad P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}} \times 100$$

Counting Techniques and Probability

$${}^{n}P_r = \frac{n!}{(n-r)!} \qquad {}^{n}C_r = \frac{n!}{(n-r)!r!}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(A \cap B) = P(A) \times P(B) \qquad P(A \cap B) = P(B) \times P(A \mid B)$$

Scatter Diagram and Correlation

$$r = \frac{n\sum XY - \left(\sum X\right)\left(\sum Y\right)}{\sqrt{\left[n\sum X^2 - \left(\sum X\right)^2\right]\left[n\sum Y^2 - \left(\sum Y\right)^2\right]}}$$

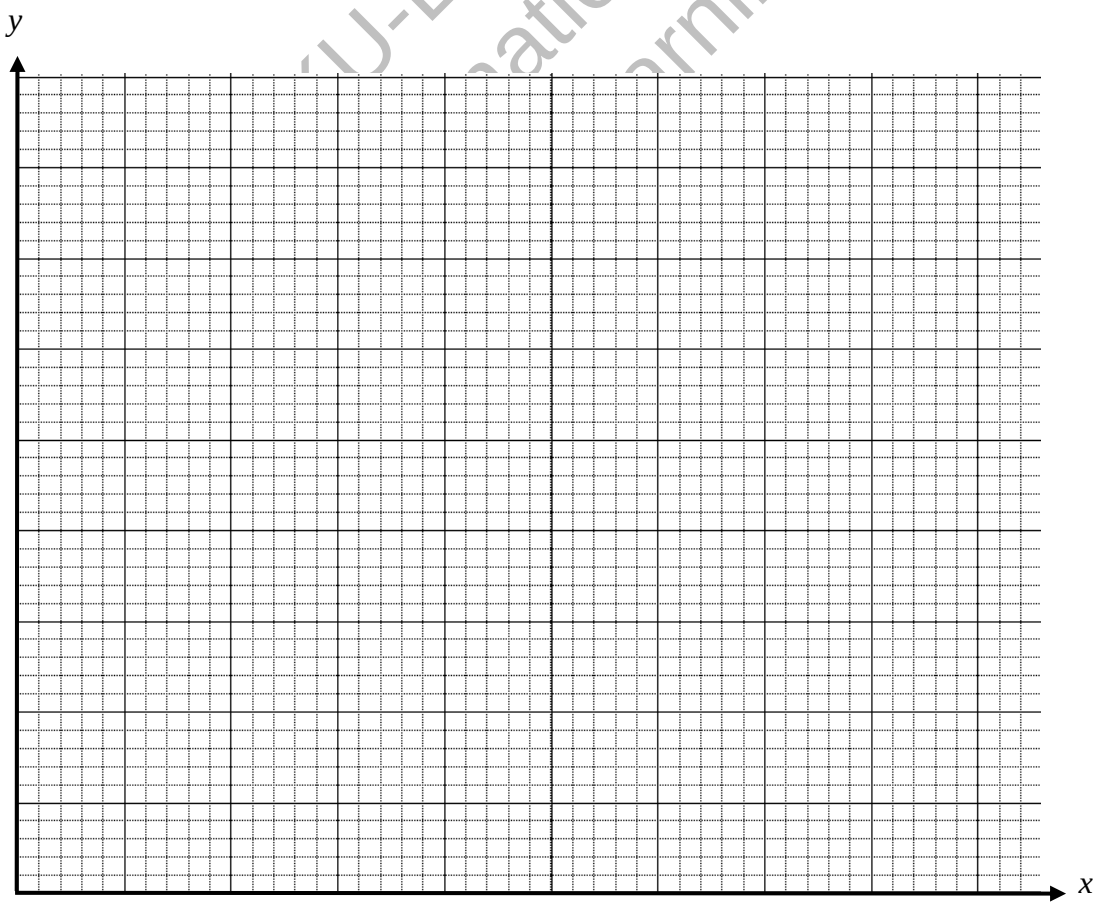
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Q.1. (Total 3 Marks)

i. Complete the given frequency distribution table. (1 Mark)

Marks	Relative Frequency	Class Mark
41.5 – 46.5	0.1	
46.5 – 51.5	0.3	
51.5 – 56.5	0.35	
56.5 – 61.5	0.2	
61.5 – 66.5	0.05	

ii. Construct the relative frequency histogram and relative frequency polygon for the data given in part (i). (2 Marks)



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(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 3 Marks)

a. The average number of ATM transactions per day of a particular bank were recorded at 10 different locations in a city.

The data were: 35, 50, 225, 50, 30, 65, 40, 55, 52 and 76.

i. Find the mean of average number of ATM transactions per day. (1 Mark)

ii. On a certain day, due to a technical fault the number of ATM transactions were decreased by half from all of the locations. After technical fault, find the new mean of average number of ATM transactions. (1 Mark)

iii. Prove the following property:

Half of the average number of ATM transactions before technical fault = Average number of ATM transactions after technical fault. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.2.)

b. Usually the flow of river is measured in cubic metre per second (m^3/s). (3 Marks)

The following are measurements of a river's annual maximum flow:

879, 261, 280, 989, 247, 302, 148, 714, 713, 094, 913, 453, 207, 530, 414

Construct a stem-leaf plot with two digit leaves.

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Q.3. (Total 3 Marks)

The IQs' of 100 students at a certain elementary level school are given in the following table.

Class Mark (*x*)	**Frequency (*f*)**			
70	4			
74	9			
78	16			
82	26			
86	45			
Total	100			

Find the variance of the students' IQ. Give your answer up to two decimal places.

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Q.4. (Total 4 Marks)

For the given data in the table, calculate Paasche's index number using base year 2010. Also interpret your result.

	2010		2015			
Commodity	**Quantity (kg)**	**Price/ kg (Rs)**	**Quantity (kg)**	**Price/ kg (Rs)**		
Wheat	100	10	120	13		
Rice	50	100	40	110		
Cotton	35	148	40	160		
Butter	12	600	10	750		
Mutton	48	500	35	600		
Total	–	–	–	–		

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(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 4 Marks)

a. A and B are two events such that $P(A) = \frac{15}{32}$, $P(B) = \frac{13}{32}$ and $P(A \text{ or } B) = \frac{23}{32}$.

i. Find $P(A \cap B)$ (1 Mark)

ii. Find $P(A' \cup B')$ (2 Marks)

iii. State whether A and B are mutually exclusive or not. Also justify your answer. (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.5.)

b. (Total 4 Marks)

Suppose two cards are selected at random without replacement from an ordinary deck of 52 playing cards in the following sequence.

A = a Queen is selected

B = a Spade is selected

Find the probability of $P(B|A)$

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Q.6. (Total 3 Marks)

The relationship between two data sets *X* and *Y* is being investigated. Both data sets contain 12 observations each. Their results are as followed:

$\sum XY = 647,\ \sum X = 80,\ \sum Y = 91,\ \sigma_x^{\,2} = 1061,\ \sigma_y^{\,2} = 467$

Compute the coefficient of correlation between two variables.

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END OF PAPER

Please use this page for rough work

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